



STEP 10 Now use an opaque cellophane tape to create a cylinder. Use that to separate the objective lens from the sensor. This distance is what gives it the magnification needed for a microscope.



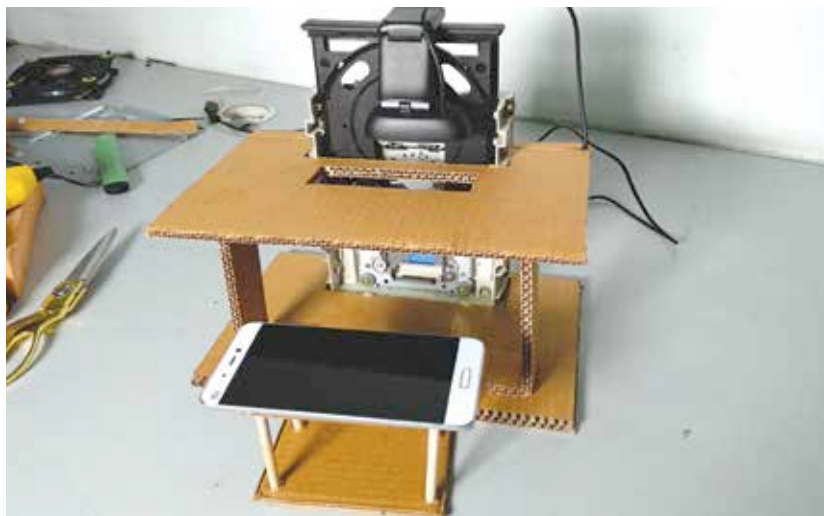
STEP 11 You now need to create a light source to illuminate whatever you place under the microscope. You can either place a led strip or use a smartphone for the backlight. We're making a mini-table here.



STEP 12 Hot glue four dowel pieces onto one rectangular sheet or cardboard. Ensure that this square sheet can easily pass between the pillars of our microscope.



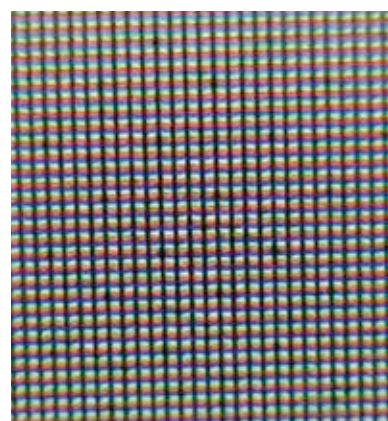
STEP 13 Hot glue a similar sized cardboard sheet onto the top to complete your mini-table.



STEP 14 Set your phone to maximum brightness and use a plain white wallpaper. Place it on the mini-table. Ensure that screen on-time is set to maximum as well.



STEP 15 Slide the mini-table with the smartphone on it under the slit. Now take a thin sheet of glass (you get them at medical stores) and put your subject on it and place it over the slit. Adjust the tray height to focus on the subject. To get more magnification, move the lens further away from the sensor. Ours was just 8 mm away from the sensor.



STEP 16 Have fun! We used the microscope to check out the sub-pixel arrangement of one of our smartphones. As you can see here, this screen is an IPS panel with a simple RGB sub-pixel arrangement.